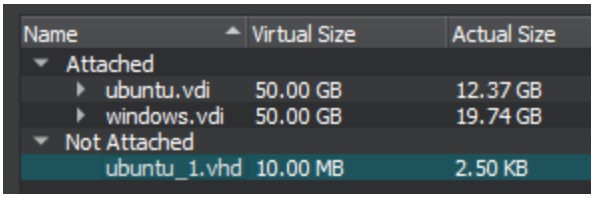
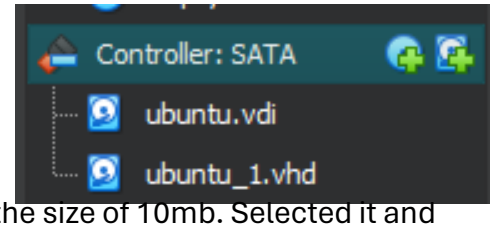
Navid Baghaei

Lab 3

TASK 1

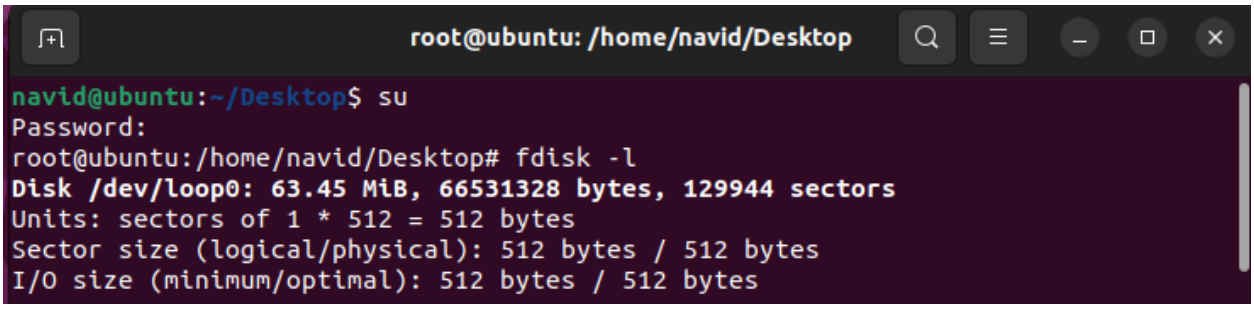
1)



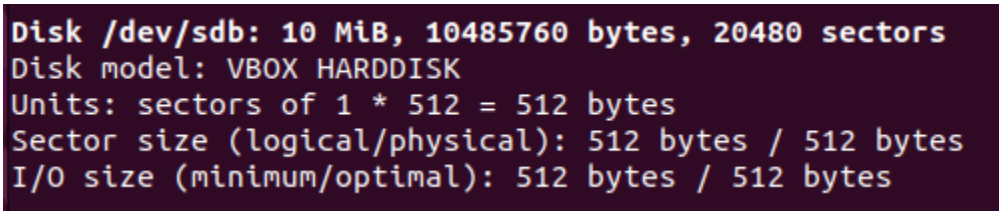


Followed the steps to create a new Hard Disk with the size of 10mb. Selected it and pressed choose.

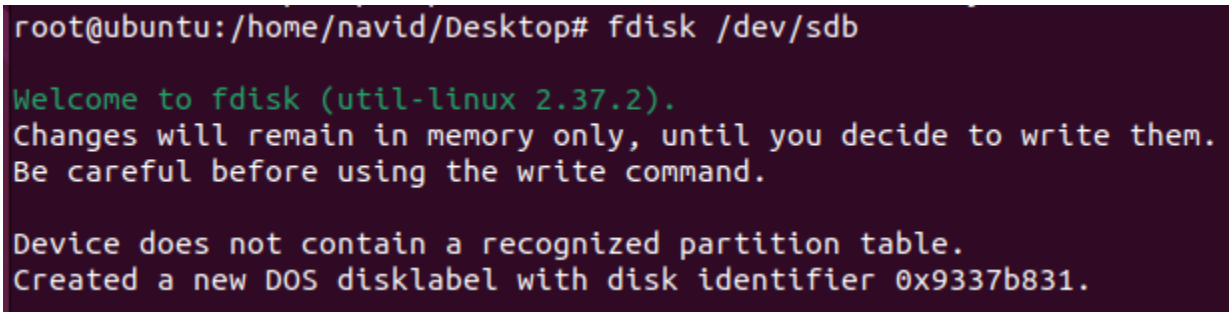
2)



Right clicked on the desktop and opened the terminal. Ran the su command to root directory. Typed “fdisk -l” to list the disks.



Here is the 10mb disk I created earlier.



Changed the directory to the new hard drive we made by running “fdisk /dev/sdb”.

```
Command (m for help): m

Help:

  DOS (MBR)
   a   toggle a bootable flag
   b   edit nested BSD disklabel
   c   toggle the dos compatibility flag

  Generic
   d   delete a partition
   F   list free unpartitioned space
   l   list known partition types
   n   add a new partition
   p   print the partition table
   t   change a partition type
   v   verify the partition table
   i   print information about a partition
```

Listed the help commands by typing “m”.

```
Command (m for help): p
Disk /dev/sdb: 10 MiB, 10485760 bytes, 20480 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x9337b831
```

Printed the partition table by typing “p”. It does not have any partitions yet.

```
Command (m for help): p
Disk /dev/sdb: 10 MiB, 10485760 bytes, 20480 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x9337b831

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p):

Using default response p.
Partition number (1-4, default 1):
First sector (2048-20479, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-20479, default 20479):

Created a new partition 1 of type 'Linux' and of size 9 MiB.
```

Created a primary partition with “n”, using the default settings.

```
Command (m for help): p
Disk /dev/sdb: 10 MiB, 10485760 bytes, 20480 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x9337b831

Device     Boot Start   End Sectors Size Id Type
/dev/sdb1        2048 20479   18432   9M 83 Linux
```

The partition now shows when running the “p” command.

```
Command (m for help): t
Selected partition 1
Hex code or alias (type L to list all): 0c
Changed type of partition 'Linux' to 'W95 FAT32 (LBA)'.

Command (m for help): l

00 Empty            24 NEC DOS          81 Minix / old Lin  bf Solaris
01 FAT12            27 Hidden NTFS Win  82 Linux swap / So  c1 DRDOS/sec (FAT-
02 XENIX root       39 Plan 9           83 Linux            c4 DRDOS/sec (FAT-
03 XENIX usr        3c PartitionMagic   84 OS/2 hidden or   c6 DRDOS/sec (FAT-
04 FAT16 <32M       40 Venix 80286      85 Linux extended   c7 Syrinx
05 Extended         41 PPC PReP Boot    86 NTFS volume set  da Non-FS data
06 FAT16            42 SFS              87 NTFS volume set  db CP/M / CTOS / .
07 HPFS/NTFS/exFAT  4d QNX4.x           88 Linux plaintext  de Dell Utility
08 AIX              4e QNX4.x 2nd part  8e Linux LVM        df BootIt
09 AIX bootable     4f QNX4.x 3rd part  93 Amoeba           e1 DOS access
0a OS/2 Boot Manag  50 OnTrack DM       94 Amoeba BBT       e3 DOS R/O
0b W95 FAT32        51 OnTrack DM6 Aux  9f BSD/OS           e4 SpeedStor
0c W95 FAT32 (LBA)  52 CP/M             a0 IBM Thinkpad hi  ea Linux extended
0e W95 FAT16 (LBA)  53 OnTrack DM6 Aux  a5 FreeBSD          eb BeOS fs
0f W95 Ext'd (LBA)  54 OnTrackDM6       a6 OpenBSD          ee GPT
```

Changed the target drive's partition to Windows 95 FAT32, using "t" and "0c" as the hex code. Ran "l" command to list all the hex to filesystem options.

```
Command (m for help): p
Disk /dev/sdb: 10 MiB, 10485760 bytes, 20480 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x9337b831

Device     Boot Start   End Sectors Size Id Type
/dev/sdb1        2048 20479   18432   9M  c W95 FAT32 (LBA)
```

Now when running the "p" command, FAT32 is now listed as the filesystem.

```
Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

root@ubuntu:/home/navid/Desktop#
```

Ran the "w" command to save the partition settings, and fdisk exited back to the terminal.

```
Disk /dev/sdb: 10 MiB, 10485760 bytes, 20480 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x9337b831

Device     Boot Start   End Sectors Size Id Type
/dev/sdb1        2048 20479   18432   9M  c W95 FAT32 (LBA)
```

When running “fdisk -l” again, we can see that the 10mb drive made earlier now shows the W95 FAT32 partition we just created.

```
root@ubuntu:/home/navid/Desktop# mkfs.msdos -vF32 /dev/sdb1
mkfs.fat 4.2 (2021-01-31)
WARNING: Number of clusters for 32 bit FAT is less then suggested minimum.
/dev/sdb1 has 255 heads and 63 sectors per track,
hidden sectors 0x0800;
logical sector size is 512,
using 0xf8 media descriptor, with 18396 sectors;
drive number 0x80;
filesystem has 2 32-bit FATs and 1 sector per cluster.
FAT size is 142 sectors, and provides 18080 clusters.
There are 32 reserved sectors.
Volume ID is 7696dfe1, no volume label.
```

Formatted the partition as FAT using “mkfs.msdos -vF32 /dev/sdb1”.

TASK 2

1)

```
Disk /dev/sda: 30 GiB, 32212254720 bytes, 62914560 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: 734F0DD2-C73F-48F7-82EA-C193AECFC4C3

Device       Start      End  Sectors  Size Type
/dev/sda1     2048     4095     2048    1M BIOS boot
/dev/sda2     4096  1054719  1050624  513M EFI System
/dev/sda3  1054720 62912511 61857792 29.5G Linux filesystem


Disk /dev/sdb: 10 MiB, 10485760 bytes, 20480 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x9337b831
```

Running “fdisk -l” again shows that sda1-3 is the host system and sdb1 is the external drive.

```
root@ubuntu:/home/navid/Desktop# mkdir /mnt/sdb1
root@ubuntu:/home/navid/Desktop# mount -t vfat /dev/sdb1 /mnt/sdb1
root@ubuntu:/home/navid/Desktop# cd /mnt/sdb1
root@ubuntu:/mnt/sdb1# ls -la
total 5
drwxr-xr-x 2 root root  512 Dec 31  1969 .
drwxr-xr-x 3 root root 4096 Feb 16 21:56 ..
root@ubuntu:/mnt/sdb1#
```

Ran the commands above to create a mount point for the external drive and mount the drive to the mount point.

```
root@ubuntu:/mnt/sdb1# ls -lah
total 8.1M
drwxr-xr-x 2 root root  512 Dec 31  1969 .
drwxr-xr-x 3 root root 4.0K Feb 16 21:56 ..
-rwxr-xr-x 1 root root 4.0M Feb 16 22:09 image_sda3aa
-rwxr-xr-x 1 root root 4.0M Feb 16 22:09 image_sda3ab
root@ubuntu:/mnt/sdb1#
```

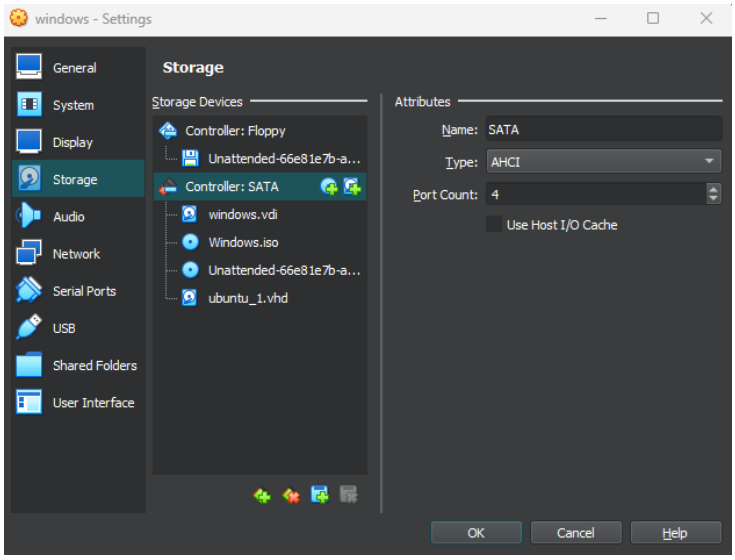
Running “ls -lah” verifies that 8mb were copied.

```
root@ubuntu:/mnt/sdb1# cd ..
root@ubuntu:/mnt# umount /dev/sdb1
root@ubuntu:/mnt#
```

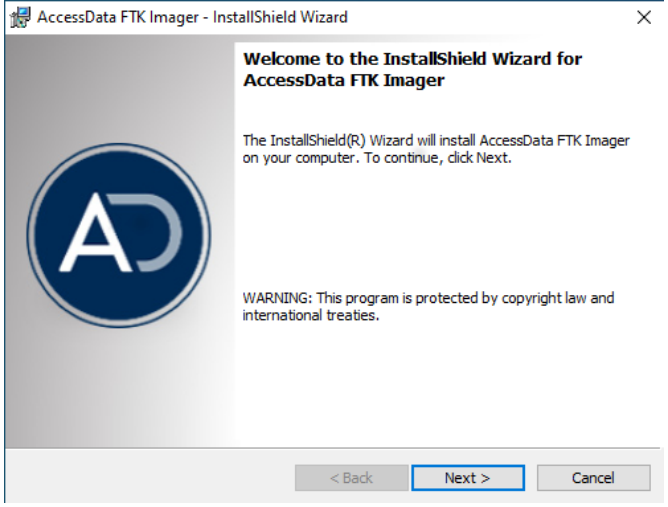
Ran “cd ..” to go back in the directory, and then ran “umount /dev/sdb1” to unmount the external disk.

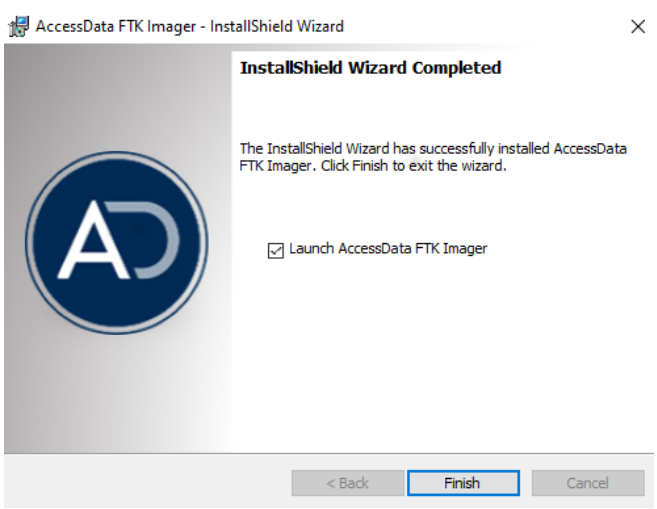
TASK 3

1)

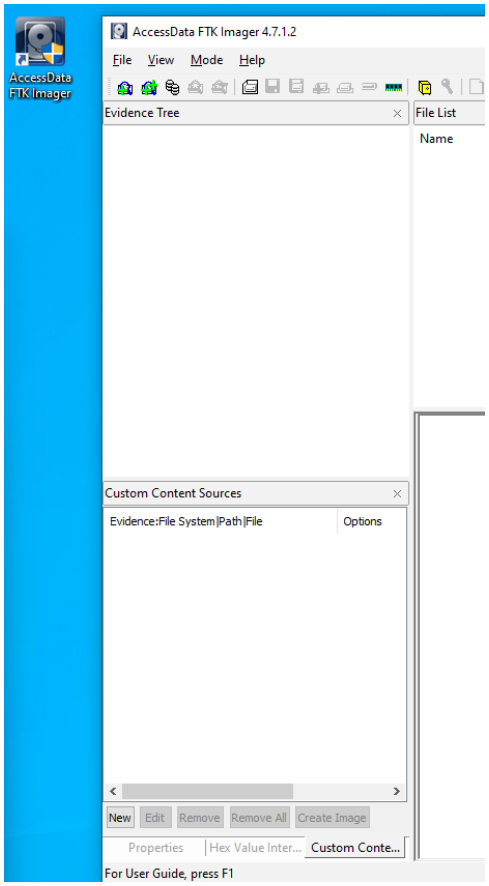


Added the 10mb hard drive we made earlier to the windows VM.



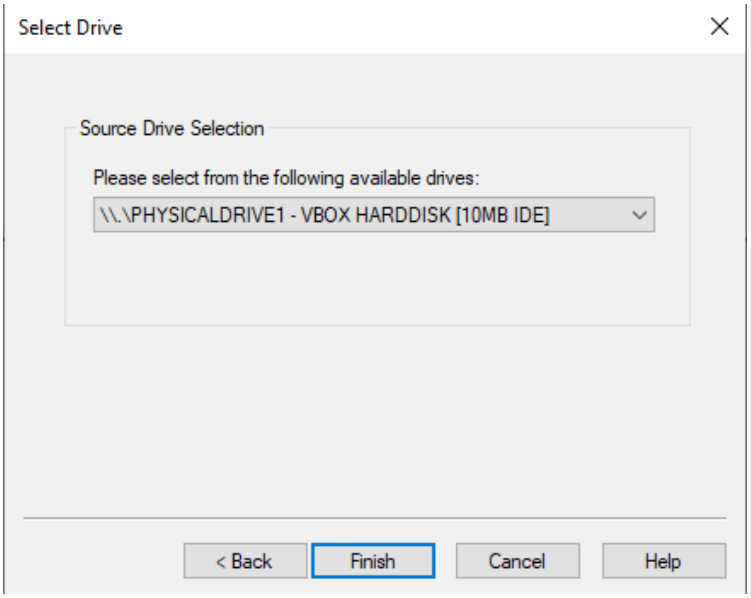


Installed FTK Imager

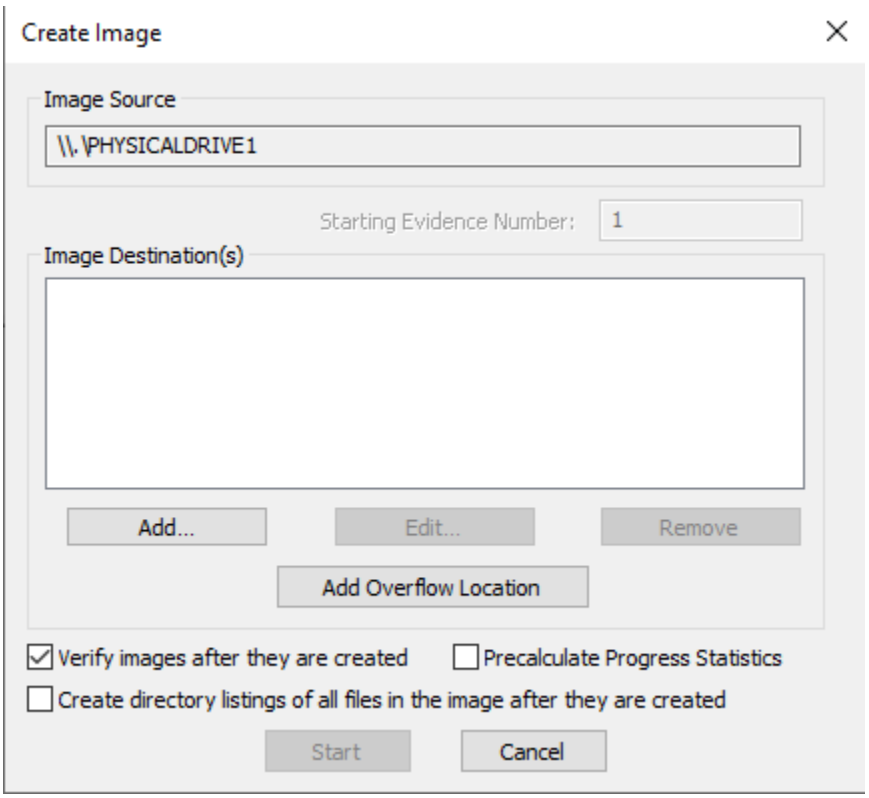


The application is on the desktop and is launched.

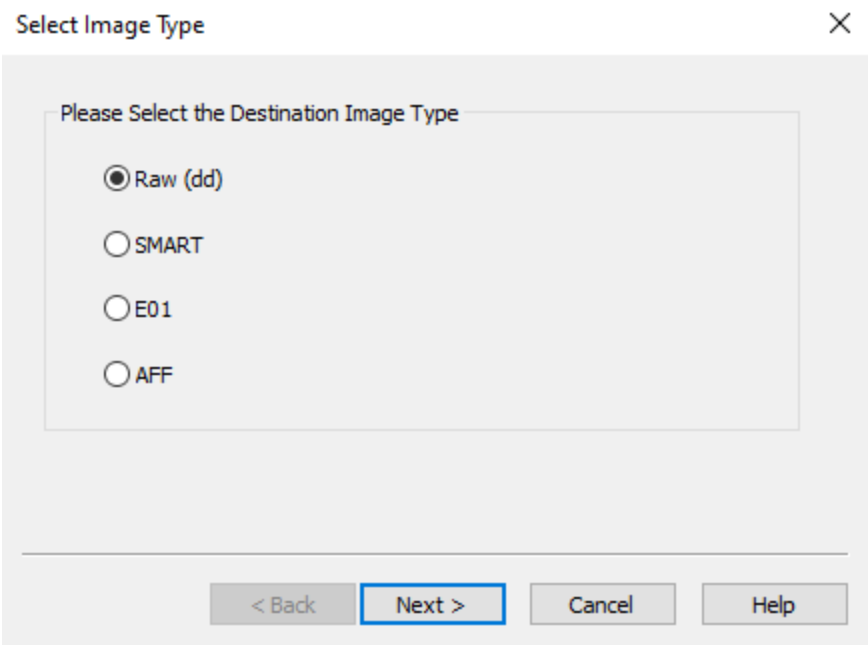
2)



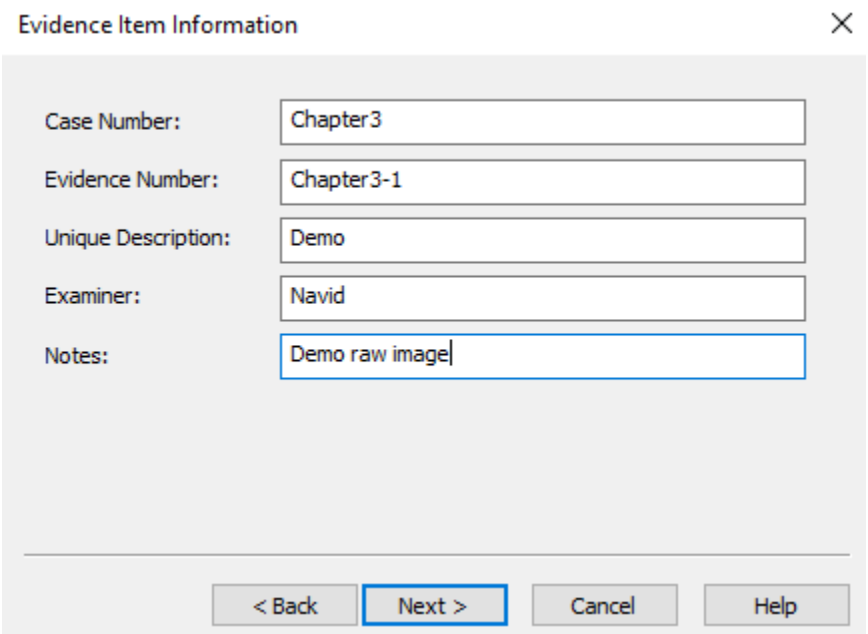
Created a disk image using the 10mb hard drive.



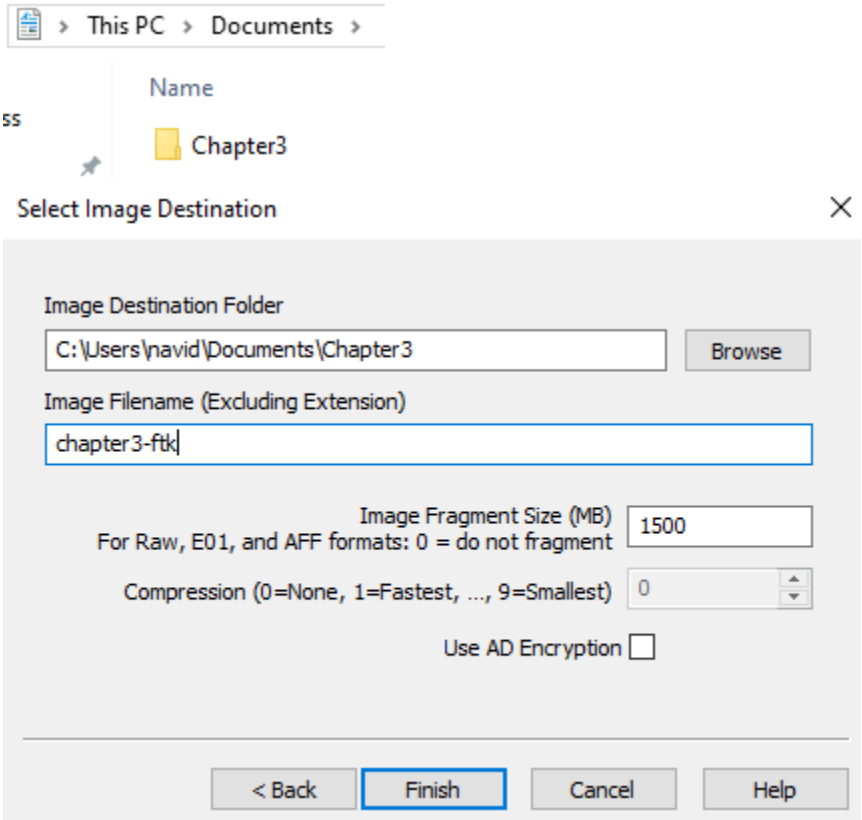
Checked the “Verify images after they are created” box and pressed add.



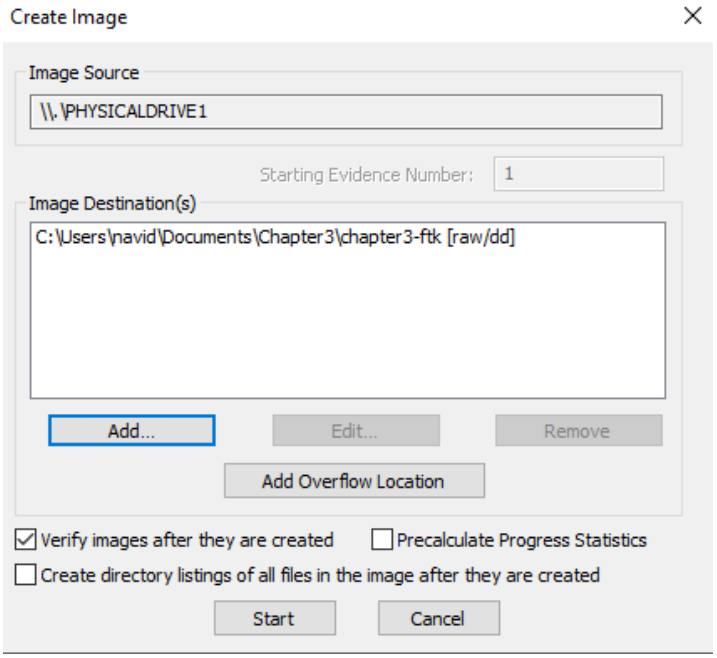
Selected Raw (dd)



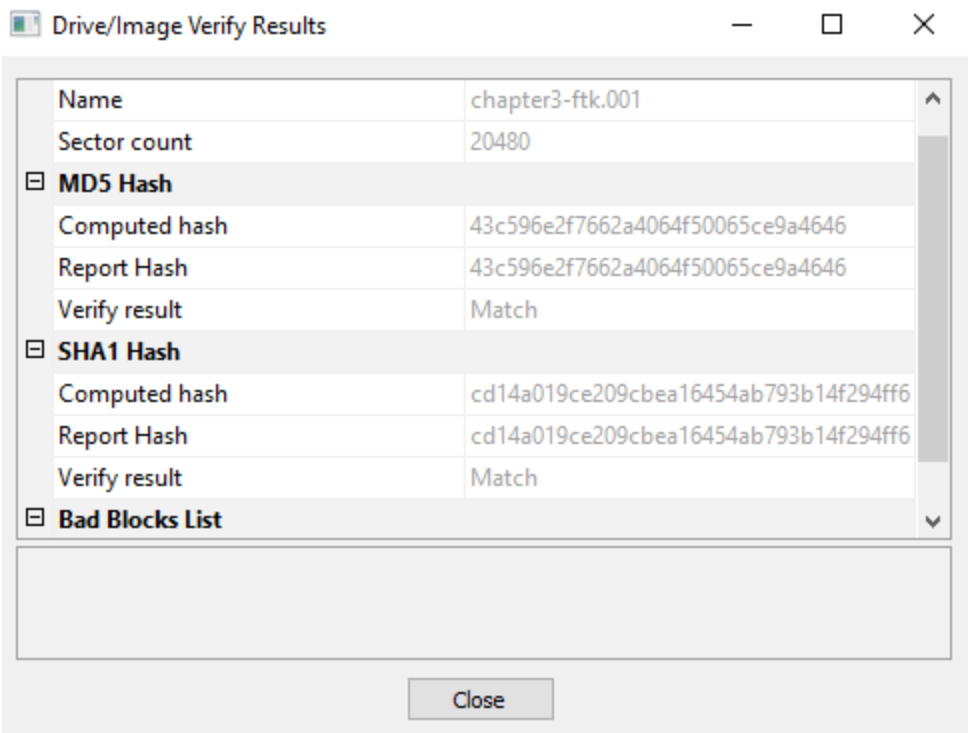
Inputted this information for the evidence item information.



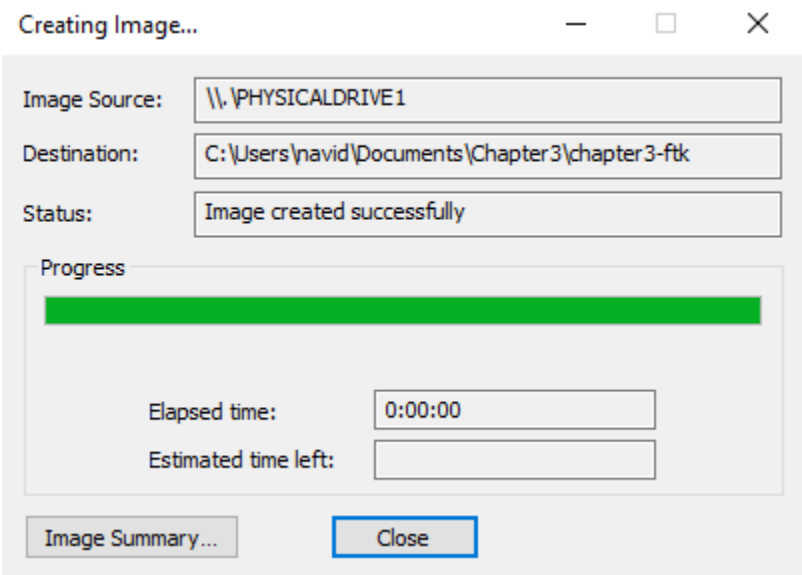
Created a folder named “Chapter3” in the VMs documents folder. Selected it as the image destination folder, and named the image filename “chapter3-ftk” and pressed finish.



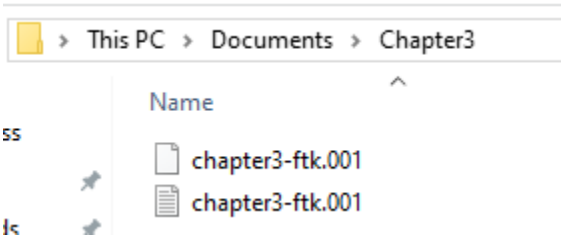
Pressed start to initiate the acquisition.



The image verify results have matches and don't show any errors.



The image was created successfully.



The image and log file have been created inside the Chapter3 folder in Documents.